

Junjie mao

Game Programmer

Summary

Game Programmer with over 2 years of game development experience. I am a highly self motivated individual that loves to learn challenging game/engine development topics. I love low level programming and building things from scratch.

Portfolios

Jason's Room, game/engine programmer

Aug. 2025 - Current

- 2D sokoban style puzzle game.
- Program and design game architecture, gameplay systems, and live code editing systems using Raylib, C++, and data oriented principles.
- Responsible for memory management using Arena Allocator, game state management using State Machine, and assets management.

Arena Shooter, game programmer

Jun. 2025 - Current

- Top-down Arena Shooter using Unity.
- Program and design gameplay system using OOP principle with C#
- Implemented sprite animation systems, enemy AI systems, texture shader effects, and game state management systems with scriptable objects, Interactive UI systems.

Vulkan Renderer, graphics programmer

Oct. 2025 - Current

- Computer graphics renderer using Vulkan graphics API with C++
- Program and design renderer architecture, rendering systems, window and input systems using GLFW, perspective pitch/yaw camera systems, and UI systems using dear ImGui.
- Implemented synchronization between cpu and gpu threads. multi model/instance rendering, texture mapping and fog effects using GLSL shaders, texture sampler and depth buffers.

Professional Experience

Open Source Contributor , RayGui — Dec. 2025

- Contributor for RayGui, an open source immediate mode gui library.
- Created a pull request that's successfully merged into master branch.
- Completes a code review in open sourced software.

Software Engineer, CinLearn— Sept. 2021 - June 2022

- Front-end development using Html, Css, and [Vue.js](#).
- Programmer on several website login widgets and systems
- Designed several website layouts

Contact

jasonmao0802@gmail.com
https://jasoncnm.github.io/portfolio_page/

SKILLS

Game Development,
Game Engine Development,
Graphics Programming,
OOP, ECS, Web Development,

Experience Tools

Vulkan, OpenGL, RenderDoc,
RAD Debugger, Visual Studio,
gdb, Unity, Unreal, Godot,
Html, Css, [Vue.js](#), emacs, Json

LANGUAGES

C++, C#, C, GLSL, Unity Shader
Graph, Lua, Python, R, Racket,
JavaScript, Html, Css, emacs
lisp.

EDUCATION

University of Waterloo Waterloo, Ontario

September 2020 - September
2024

Bachelor of Math, Honours
Computer Science, and Joint
Honours Combinatorics and
Optimization